

MOHAMMED ISLAM - SC CLEARED

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PROFILE

Platform engineer with a strong background in enterprise-level Kubernetes support, Kubernetes release management, observability & monitoring, & feature development.

EDUCATION

University of Warwick

September 2013 - August 2017

Department of Engineering - MEng Systems Engineering

CERTIFICATIONS

SKILLS

Soft Skills

Project Management, Team Work, Peer Mentoring, On-site & Remote Working, Release Management, Incident & Problem Management, Disaster Recovery Planning, System Architecture

Observability & Monitoring

Cloudwatch, Datadog, Pagerduty, Elasticsearch, Opensearch, Fluentd, Logstash, Beats, Metrics-server, Prometheus, Grafana, Thanos, Loki, Mimir, Tempo

Scripting, Programming & Other Languages

Python, Shell (Bash)

Configuration Management & DevOps Tools

Confluence, Service Now, JIRA, Trello, Jenkins, Gitlab, Git, Twistlock, Vault, NGINX, AWS, Kubernetes, Helm, Gatekeeper, OPA, Conftest, Kustomize, Terraform, Terragrunt, Docker, GCP, Azure, AKS, KEDA, Karpenter

EXPERIENCE

Phillip Morris International

June 2024 - Present

Senior Platform Engineer - Kubernetes & Cloud Infrastructure

- Architected enterprise-scale Kubernetes platform using Cluster API with integrated day-2 operations tooling, providing 50+ development teams with standardized, secure cloud infrastructure
- Led end-to-end PagerDuty to OpsGenie migration in 6 months, ensuring 100% feature parity while coordinating 20+ stakeholder teams and delivering \$150K in annual cost savings
- Developed Node.js middleware application integrating ServiceNow incident data with real-time market-location-based status updates, enabling executive leadership monitoring across global operations
- Designed declarative infrastructure provisioning platform using Terraform modules and custom Golang tooling, enabling complete cloud environment deployment through single YAML configurations
- Implemented OIDC federated authentication for service repositories with External Secrets Operator integration, eliminating manual secret management across 50+ microservices
- Engineered automated disaster recovery testing framework using Point-in-Time Recovery (PITR), reducing manual testing effort by 80% and ensuring 4-hour recovery objectives

HMRC

January 2023 - March 2024

Site Reliability Engineer - Platform Squad (Kubernetes Specialist)

- Delivering EKS as a Platform with integrated day-2 tooling to customers to achieve a unified, pen-tested secure cloud platform across the organisation
- Implemented standardised compliance testing through Open Policy Agent & Conftest. Moving from multiple runners to a single multi-tenant deployment runner agent against AWS infrastructure, defining permissions based on standardised YAML per team and enforcing against Terraform JSON. This saved runner costs by 70% and reduced engineering hours spent
- Architected a series of Lambdas to populate and distribute SSH keys with automated rotation within a 60 day period
- Created DynamoDB tables to dynamically store AMI status for different accounts
- Designed automation to evacuate EKS nodes from selected AZs, increasing organisation DR capabilities
- Designed Python program leveraging Github API and Jinja2 templating to automate release note generation across different internal EKS releases, bringing down release note generation time from 4-6 hours manually, to 15 seconds

William Hill

June 2022 - January 2023

Site Reliability Engineer - Kubernetes Platform

- Worked on deploying Prometheus as a vended repository for consumers to utilise a satellite Prometheus with a Thanos sidecar within their EKS cluster to unify platform observability.
- Deployed Thanos chart, alongside additional components such as ruler & compactor with associated bucket, access points and role infra to enable for long term object storage of Prometheus metrics.
- Leveraging of Prometheus kube state metrics metadata to alert based on team labelling allowing custom receivers to fire.
- Improved Opensearch performance capabilities by performance testing and optimising JVM heap size to increase garbage collection efficiency.
- Implemented custom Prometheus exporters to generate metrics for AWS services into a single monitoring system, allowing Product Owners to streamline platform service offerings.
- Monitored and maintained AWS services, including EC2 instances, S3 buckets, and RDS databases, ensuring high availability through Cloudwatch pushing metrics to Prometheus Pushgateway.

Nationwide Building Society

September 2017 - June 2022

Site Reliability Engineer - Enterprise Kubernetes Team

2020 - June 2022

- Automated infrastructure management and deployment using Terraform and Terragrunt for AWS Cloudwatch & RDS modules for onprem developers via Python Jinja2 wrappers, easing deployment use for customers and eliminating team support tickets by 50% thanks to new developer self serve capabilities
- Supporting developer teams with microservice Github Actions to deploy to multiple platforms and enforce pull request policies such as tagging and linting
- Adjustment of pipelines to automatically shut down lab clusters over evenings and auto destroy after a 2 weeks, saving costs by 85%
- Designed multi-stage Buildkite pipeline, for cluster build & test stages using Make targets & 3-musketeers pattern

Platform Engineer - Observability & Monitoring Team

2018 - 2020

- Development of monitoring platform, deploying Logstash architecture alongside Beats family to enable centralized application logging within Elasticsearch and visualisation within Kibana to achieve a unified monitoring platform spanned 40+ teams

- Created Grafana dashboard pipeline allowing consumers to self serve dashboards to team specific custom folders via Helm templating and automatically promote to production via pipeline scripting, reducing dashboard support tickets by 90%

Platform Engineer - Digital Transformation

September 2017 - 2018

OVO Energy & Warwick Manufacturing Group

July 2016 – June 2017

Research Analyst

- Modelled the degradation of Nissan Leaf batteries through lab testing & MATLAB